Jessica — 37/40 (Q1 12/12 · Q2 10/13 · Q3 15/15)

南京外国语学校
NANJING FOREIGN LANGUAGE SCHOOL
International Centre

STUDENT	Jessica Gao YueMing 3941
TEACHER	Gong Qiushi

CIE Semester 1 Midterm Examination 2026 - 2027

Paper 1&2 Combined

AS Computer science

60 Minutes

No additional materials are needed

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use red ink, staples, paper clips, highlighters, glue or correction fluid.
The exam paper contains 3 structured questions.
Answer **all** your questions on the **exam booklet.**

Q1	**/ 12**
Q2	**/ 13**
Q3	**/ 15**
Total	**/ 40**

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of 8 printed pages.

Cambridge Assessment International Education

[TURN OVER]

1. A gym uses a program to store information about its members.

(a) State the most appropriate **data type** for each item of data.

Item of data	Data type
the membership number — a whole number between 1 and 9999	INTEGER
the monthly fee — for example 299.50	REAL
whether the member has paid for this month	BOOLEAN
the membership grade — a single letter, A, B, C or D	CHAR

[4]

(b) The following pseudocode is run.

```
DECLARE Count : INTEGER
DECLARE Total : INTEGER

Count <- 1
Total <- 0

WHILE Count <= 4
   Total <- Total + Count * 2
   Count <- Count + 1
ENDWHILE

OUTPUT Total
OUTPUT Count
```

Complete the trace table below.

Count	Count <= 4 ?	Total
1	TRUE	2
2	TRUE	6
3	TRUE	12
4	TRUE	20
5	FALSE	20

Output:

Total = 20

Count = 5 [4]

(c) A programmer writes the following pseudocode. It is intended to output the numbers 10, 9, 8, 7, 6, 5, 4, 3, 2, 1.

```
DECLARE N : INTEGER

N <- 10

WHILE N >= 0
   OUTPUT N
   N <- N + 1
ENDWHILE
```

(i) State the error in this pseudocode and explain what happens when it is run.

N ← N+1 is wrong. The pseudocode will output numbers followed larger than 10. The loop will never end. And N can't be equal to 0, because the output doesn't contain 0 [2]

(ii) Rewrite the pseudocode so that it outputs the numbers as intended.

```
DECLARE N: INTEGER
N ← 10
WHILE N > 0 DO
    OUTPUT N
    N ← N - 1
ENDWHILE
```

[2]

2. The following pseudocode defines a function.

```
FUNCTION Calc(A : INTEGER, B : INTEGER) RETURNS INTEGER
   IF A > B THEN
      RETURN A * 2
   ELSE
      RETURN B + A
   ENDIF
ENDFUNCTION
```

(a) State the value returned by each of the following calls.

Call	Value returned
`Calc(7, 3)`	14
`Calc(2, 9)`	11
`Calc(5, 5)`	10

[3]

(b) Write pseudocode for a function `DigitTotal` that takes an `INTEGER` parameter and **returns** the sum of its digits. For example, `DigitTotal(472)` returns 13.

```
FUNCTION DigitTotal(num:INTEGER) RETURNS INTEGER
    DECLARE tens,cnt,index : INTEGER
    cnt ← 0
    index ← num
    tens ← 1
    WHILE tens <= num DO
        index ← index ÷ 10
        tens ← tens * 10
        cnt ← cnt + index MOD 10
        index ← index DIV 10
    END WHILE
    RETURN cnt
ENDFUNCTION
```

[5]

(c) Explain the difference between a **function** and a **procedure**.

A function has a return value while a procedure doesn't. And a procedure can be directly used but a function needs to be used by "CALL".

[2]

(d) Explain what is meant by **modularization** and give **one** benefit of using it.

It means to find the value of the remainder after a number is ~~divided~~ divided by another number. It is ~~used~~ useful for finding the last digit of an integer by ~~dividing~~ MOD 10 ~~by 10~~ ~~MOD 10~~.

[3]

Q2(d) 0/3
This answers MODULO (the remainder), not modularisation. Modularisation = breaking a program into smaller, independent modules (functions / procedures), each doing one well-defined task. Benefit: modules can be written and tested separately, reused, and the program is easier to read and maintain.

3. A school stores the scores of 30 students in a one-dimensional array `Scores`.

(a) Write pseudocode to declare the array `Scores`.

```
DECLARE Scores : ARRAY[1:30] OF INTEGER
```

[2]

(b) The array `Scores` has been filled with 30 values. Write pseudocode to output the **largest** value in `Scores` and the **index** at which it is found.

```
~~DECLARE index, maxi: INTEGER~~
DECLARE index, maxi, pos : INTEGER
index ← 1
maxi ← -1
pos ← 1
FOR index ← 1 TO 30
    IF Scores[index] > maxi THEN
        maxi ← Scores[index]
        pos ← index
    ENDIF
NEXT index
OUTPUT maxi, pos
```

[5]

(c) Write pseudocode for a function `CountPass` that takes the array `Scores` and an integer `PassMark` as parameters and **returns** the number of values in `Scores` that are **greater than or equal to** `PassMark`.

```
FUNCTION CountPass ( Scores : ARRAY[1:30] OF INTEGER , PassMark :
                    INTEGER) RETURNS INTEGER
DECLARE cnt , index : INTEGER
cnt ← 0
index ← 1
FOR index ← 1 TO 30
    IF Scores [index] >= PassMark THEN
        cnt ← cnt + 1
    ENDIF
NEXT index
RETURN cnt
ENDFUNCTION
```

[5]

(d) The school also stores the marks for three subjects in a two-dimensional array `Marks`. `Marks` has **5 rows** and **3 columns**.

(i) Write pseudocode to declare the array `Marks`.

```
DECLARE Marks : ARRAY [1:5 , 1:3] OF INTEGER
```

[1]

(ii) Write pseudocode to output **every** value in `Marks`.

```
DECLARE rownum , colnum : INTEGER
rownum ← 1
colnum ← 1
FOR rownum ← 1 TO 5
    FOR colnum ← 1 TO 3
        OUTPUT Marks [ rownum , colnum ]
    NEXT colnum
NEXT rownum
```

[2]

End of the exam.

Use following pages for scratch paper.